Quantum Computation and Information

Lecture 4: Quantum Complexity Theory

- classical complexity
- quantum complexity

solvable problem

- which problems will we be able to solve in practice?

solvable problem

- which problems will we be able to solve in practice?
- a working definition: those with polynomial-time algorithms
  - mathematically, the definition is broad and robust: closed closed under addition, multiplication, composition; huge gulf between polynomials and exponentials
  - practically, polynomial-time algorithms scale to huge problems, e.g., the time increases by a small constant when the input size is doubled.
- solvable problems (classically): shortest path, min cut, 2-satisfiability, primality testing, ...
- problems (believed be to) not solvable (classically): longest path, max cut, 3-satisfiability, factoring, ...
- problems provably requiring exponential-time: halting problem (given a constant-size program, does it halt in at most k steps?), ...

complexity theory

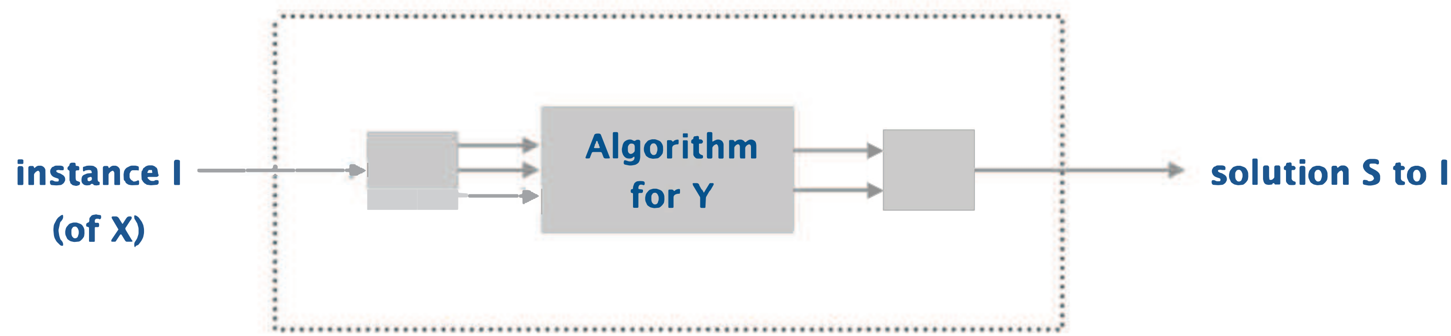
- classify hardness of problems in terms of resources required to solve algorithmically large instances
  - those that can be solved in polynomial time versus those that cannot
- computational efficiency measured by the scaling of resources (time/memory) of the algorithm with input size of an instance
  - size of an instance is number of bit required to specify it
- robust classification up to a polynomial overhead
- classical 70's and quantum late 90's

polynomial-time reduction

problem X polynomial-time reduces to problem Y if arbitrary instances of problem X can be solved using:

- polynomial number of basic computational steps, plus

- polynomial number of calls to oracle that solves problem Y



Algorithm for X

- notation: $X \leq_P Y$

polynomial-time reduction

- design algorithms: If $X \leq_P Y$, Y polynomial-time solvable implies X polynomial-time solvable (X is no harder than Y)

- establish intractability: If $X \leq_P Y$ and X not polynomial-time solvable, Y cannot be solved in polynomial time (Y is at least as hard as X)

- establish equivalence: X and Y are equivalent If both $X \leq_P Y$ and $Y \leq_P X$, denoted by $X \equiv_P Y$

- reduction classifies problems according to relative difficulty up to a polynomial overhead

different types of problems

- decision problems: given a question, an input and a threshold w, determine the existence of a solution at least as good as w, answering YES if such a solution exists and NO otherwise

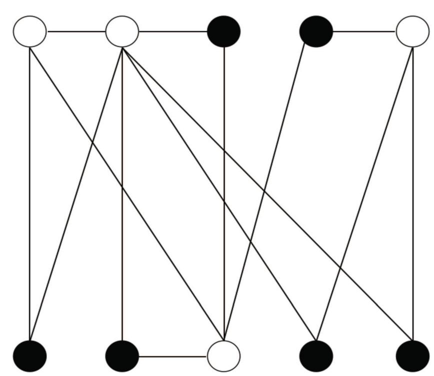
- search problems: given a question, an input and a threshold w, find a solution at least as good as w

- optimization problems: given a question and an input, find the best solution

independent set

independent-set: given a graph $G = (V, E)$ and an integer k, is there a subset of vertices $S \subseteq V$ such that $|S| \geq k$, and for each edge at most one of its endpoints is in S?

- decision problem: does there exist an independent set of size ≥ 6?
- search problem: find an independent set of ≥ 6
- optimization problem: find an independent set of maximum size



an independent set of size 6 (black vertices)

different types of problems

- decision problem: does there exist an independent set of size ≥ 6?
- search problem: find an independent set of ≥ 6
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which one is harder?

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which one is harder?

decision $\equiv_P$ search $\equiv_P$ optimzization

- holds for many problems
- proved for NP-complete problems

will describe computational complexity in terms of decision problems

solving versus verifying

three basic complexity classes

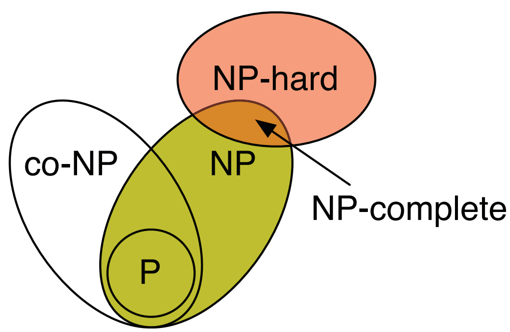
- P: decision problems that can be solved in polynomial time
- NP (nondeterministic polynomial): decision problems such that if the answer is YES, then there exists a witness (or proof) of that fact that can be checked in polynomial time
- co-NP: decision problems such that if the answer is NO, then there exists a witness (or proof) of that fact that can be checked in polynomial time

P versus NP

- P$\subseteq$NP
  - verify YES by providing a solution in polynomial time
  - to be NP, no need to find a solution, just verify a solution
- P$\subseteq$co-NP
  - algorithm for search problem provides a proof
- P$\neq$NP (believed but not proved; the most important open question in theoretical CS)
  - finding a solution is often difficult and verifying a solution seems easier, but we do no know
- NP$\neq$co-NP (believed but not proved)
  - finding a needle hiding in a haystack is hard, but showing no needle in even harder

- NP-hard: a problem such that every NP problem polynomial-time reduces to it
  - a polynomial-time algorithm for an NP-hard problem implies polynomial-time algorithm for all NP problems, i.e., P= NP
  - at least as hard as any problem in NP

- NP-complete: an NP problem that is NP-hard
  - an NP problem such that every NP problem polynomial-time reduces to it
  - NP-complete problem captures the worst case complexity of all problem in NP (hardest problems in NP)



solving versus verifying

Cook-Levin Theorem: Boolean Satisfiability is NP-complete.

- decide if a string of boolean variables, chained together with ANDs, ORs and NOTs, have any setting that yields a YES

$$\Phi = (\bar{x_1} \vee x_2 \vee x_3) \wedge (x_1 \vee \bar{x_2} \vee x_3) \wedge (\bar{x_1} \vee x_2 \vee x_4)$$

- to show a new problem Q is NP-complete, simply show $SAT \leq_P Q$

satisfiability

- literal: a Boolean variable x_i or its negation $\bar{x_i}$
- clause: a disjunction of literals, e.g., $C_j = x_1 \vee \bar{x_2} \vee x_3$
- conjunctive normal form (CNF): a propositional formula that is a conjunction of clauses, e.g., $\Phi = C_1 \wedge C_2 \wedge C_3 \wedge C_4$
- SAT: given a CNF formula Φ, does it have a satisfying truth assignment?
- 3-SAT: SAT where each clause contains exactly 3 literals

$$\Phi = (\bar{x_1} \vee x_2 \vee x_3) \wedge (x_1 \vee \bar{x_2} \vee x_3) \wedge (\bar{x_1} \vee x_2 \vee x_4)$$

YES instance: $x_1 =$true, $x_2 =$true, $x_3 =$false, $x_4 =$false

- 3-SAT is NP-Complete

recap

classical complexity classes

- P: decision problems that can be solved in polynomial time
- NP (nondeterministic polynomial): decision problems such that if the answer is YES, then there exists a witness (or proof) of that fact that can be checked in polynomial time
- NP-hard: a problem such that every NP problem polynomial-time reduces to it
- NP-complete: an NP problem that is NP-hard

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$$\Phi = (\bar{x_1} \vee x_2 \vee x_3) \wedge (x_1 \vee \bar{x_2} \vee x_3) \wedge (\bar{x_1} \vee x_2 \vee x_4)$$

YES instance: x_1 =true, x_2 =true, x_3 =false, x_4 =false

- 3-SAT is NP-Complete

quantum complexity classes

- BQP (bounded-error quantum polynomial time): a decision problem D is in BQP if there exists a polynomial-time quantum algorithm that decides D with a bounded error rate (say, at most $1/3$)
  - quantum generalization of P; quantumly easy
  - error is allowed because of measurement being probabilistic
  - bounded error rate is required to obtain any desired confidence by repetition
  - example: factoring
- P$\subseteq$BQP
  - any classical circuit can be implemented with Toffoli gates with a polynomial overhead
- NP$\not\subseteq$BQP (believed but no proved)
  - quantum computer cannot solve NP-complete problem in polynomial time

quantum complexity classes

- QMA (quantum Merlin Arthur): a decision problem D is in QMA if a ‘yes‘ instance can be verified by a polynomial-time quantum computation with bounded error rate (say, at most $1/3$)
  - Merlin, with unbounded computational resources, provide s a proof/certificate to Arthur who will verify it
  - quantum generalization of NP; quantumly hard
- QMA-hard: a problem such that every QMA problem polynomial-time reduces to it
- QMA-complete: a QMA problem that is QMA-hard
  - QMA-complete problem captures the worst case complexity of all QMA problems
- example: local Hamiltonian problem

quantum complexity classes

local Hamiltonian

- Hamiltonian is self-adjoint and governs evolution of (closed) quantum system (Schrödinger equation)

$$\frac{d}{dt}|\psi(t)\rangle = -iH|\psi(t)\rangle$$

  - eigenvalues are called energies in physics
  - ground state (i.e., state with lowest energy) is of particular interest

- k-local Hamiltonian: an n-qubit Hamiltonian $H = \sum_i H_i$ is k-local if each H_i acts on at most k qubits, where k is a constant and $\|H_i\| \leq h$
  - spatial locality: stronger conception of locality from physics where there is no long-range interaction
  - e.g., $H = -\sum_{i=0}^{n-2} J_{i,i+1} Z_i Z_{i+1}$ for 1-D lattice

- 3-SAT as 3-local classical Hamiltonian

$$H(x) = \sum_c H_c(x_1^c, x_2^c, x_3^c)$$

where $x_i^c \in \{0, 1\}$ and

$$H_c(x_1^c, x_2^c, x_3^c) = \begin{cases} 0 & \text{if clause } C(x_1^c, x_2^c, x_3^c) \text{ is true} \\ 1 & \text{otherwise} \end{cases}$$

  – $\min_x H(x) = 0$ iff there exists an YES assignment and $\min_x H(x) \geq 1$ otherwise
  – finding ground state energy of 3-local Hamiltonian is NP-hard
  – finding ground state energy of 3-local Hamiltonian is hard for quantum computer too, unless NP⊆BQP

- 2-local classical Hamiltonian

$$H(x) = \sum_c H_c(x_0^c, x_1^c)$$

  - finding ground state energy of 2-local Hamiltonian is NP-hard
    * MAX 2-SAT (finding the maximum number of satisfies clauses) is NP-hard, although 2-SAT is easy
  - spatial locality does not lead to loss of hardness, e.g., 2-D Ising spin-glass

$$H = -\sum_{\langle i,j \rangle} J_{ij} Z_i Z_j$$

    * $Z_i \in \{-1, 1\}$ and $\langle i, j \rangle$ neighbors on 2-D lattice
    * $J_{ij} \in \{-1, 1\}$; $J_{ij} = 1$ favoring alignment ($Z_i = Z_j$) and $J_{ij} = -1$ anti-alignment ($Z_i \neq Z_j$)
    * negative J_{ij} generates frustration where not all edges have minimum energies

- quantum case might be even harder
  - ground state can be highly entangled, with no low-complexity classical description

quantum complexity classes

quantum k-local Hamiltonian problem

- k-local Hamiltonian: an n-qubit Hamiltonian $H = \sum_i H_i$ is k-local if each H_i acts on at most k qubits, where k is a constant and $\|H_i\| \leq h$
- local Hamiltonian problem: given E_l and E_h with $E_h - E_l > \frac{1}{n^{O(1)}}$, it is promised that ground state energy E_0 of H satisfies either $E_0 \leq E_l$ or $E_0 > E_h$. decide if $E_0 \leq E_l$ or $E_0 > E_h$
  - to determine ground state energy up to $\frac{1}{n^{O(1)}}$ accuracy
  - a QMA problem
- Theorem (KSV02): local Hamiltonian problem is QMA-complete
  - spatially 2-local Hamiltonian problem in 2-D is QMA-complete